

Conclusion: q . But, since p is false, we cannot conclude anything regarding the truth value of q .

We note q is false: since $(\sqrt{2})^2 = 2 < (\frac{13}{2})^2 = \frac{9}{4} = 2.25$

Table 1: Rules of Inference

<u>Rules of Inference</u>	<u>Tautology</u>	<u>Name</u>
1) $\frac{p}{p \rightarrow q}$ $\therefore q$	$(p \wedge (p \rightarrow q)) \rightarrow q$	Modus ponens
2) $\frac{\neg q, p \rightarrow q}{\therefore \neg p}$	$(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$	Modus tollens
3) $\frac{p \rightarrow q, q \rightarrow r}{\therefore p \rightarrow r}$	$((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$	Hypothetical syllogism
4) $\frac{p \vee q, \neg p}{\therefore q}$	$((p \vee q) \wedge \neg p) \rightarrow q$	Disjunctive syllogism
5) $\frac{p}{\therefore p \vee q}$	$p \rightarrow (p \vee q)$	Addition
6) $\frac{p \wedge q}{\therefore p}$	$(p \wedge q) \rightarrow p$	Simplification
7) $\frac{p, q}{\therefore p \wedge q}$	$((p) \wedge (q)) \rightarrow (p \wedge q)$	Conjunction
8) $\frac{p \vee q, \neg p \vee r}{\therefore q \vee r}$	$((p \vee q) \wedge (\neg p \vee r)) \rightarrow (q \vee r)$	Resolution

We have already verified that these rules of inference are valid argument forms in Ex-1.3 Prob 13-16, 25, 33 and 34.